

$$2a) V(x) = \frac{V_0}{d^4} (x^4 - 2x^2d^2 + d^4) = \frac{V_0}{d^4} (x^2 - d^2)^2$$

$$\begin{aligned} V'(x) &= \frac{V_0}{d^4} \frac{d(x^2 - d^2)^2}{dx} = \frac{V_0}{d^4} (2x)(2(x^2 - d^2)) \\ &= \frac{4V_0}{d^4} x(x^2 - d^2) = 0 \quad x=0, \quad x=+d, \quad x=-d \\ &\quad x^2 - d^2 = 0 \end{aligned}$$

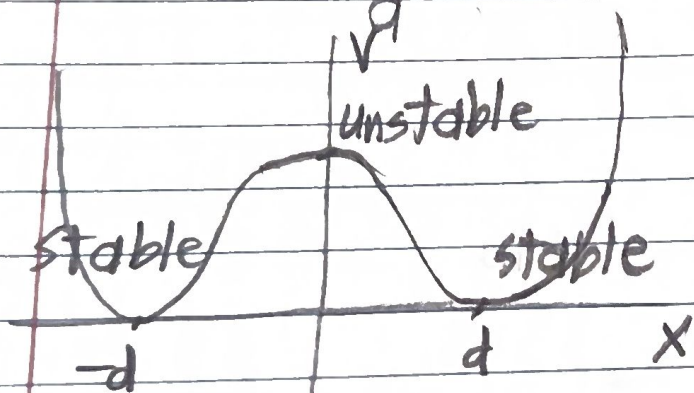
$$V''(x) = \frac{4V_0}{d^4} (x^2 - d^2 + x(2x)) = \frac{4V_0}{d^4} (3x^2 - d^2)$$

$$V''(0) = \frac{4V_0}{d^4} (0 - d^2) = -\frac{4V_0}{d^2} < 0 \quad \text{unstable}$$

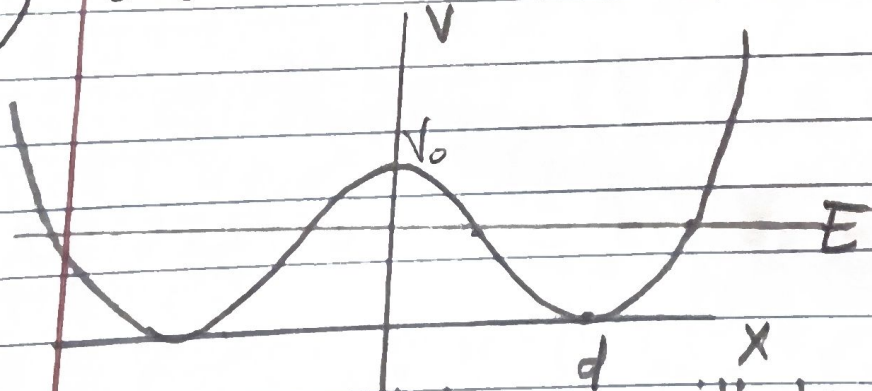
$$V''(d) = \frac{4V_0}{d^4} (3d^2 - d^2) = \frac{8V_0d^2}{d^4} = \frac{8V_0}{d^2} > 0 \quad \text{stable}$$

$$V''(-d) = \frac{8V_0}{d^2} > 0 \quad \text{stable}$$

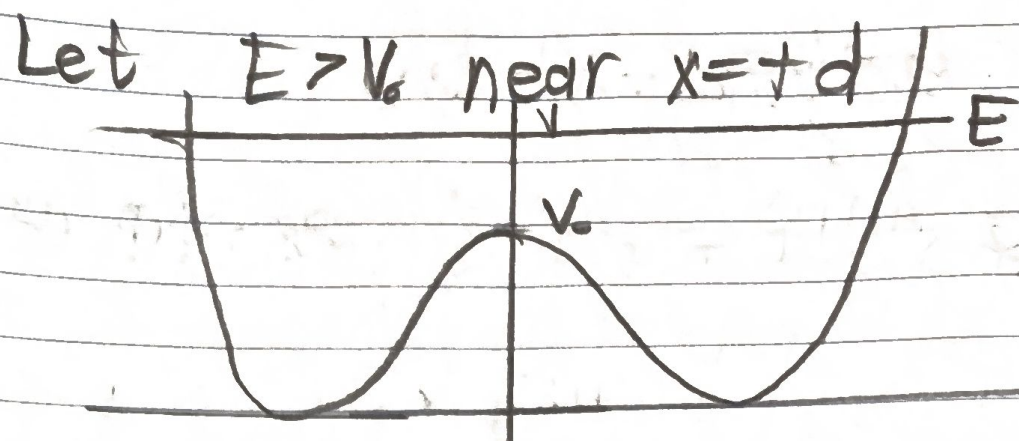
$$V(0) = \frac{V_0}{d^4} (-d^2)^2 = V_0$$



2b) Let $E < V_0$ near $x = +d$



The particle oscillates around $x = +d$. It can only reach the points in x where $V(x) = E$ but can go beyond that range.



The particle has enough energy to cross the barrier of height V_0 . It can oscillate between the points in x where $V(x) = E$

$$2c) E = \frac{1}{2} m v_0^2 + V(2d) = \frac{V_0}{d^4} ((2d)^2 - d^2)^2 = \frac{V_0}{d^4} (3d^2)^2$$

$$= 9V_0$$

$$\frac{1}{2} m v^2 + V(d) = 9V_0$$

$$\frac{1}{2} m v^2 = 9V_0 - \frac{V_0}{d^4} (d^2 - d^2)^2 = 9V_0$$

$$v = \sqrt{\frac{18V_0}{m}} = 3\sqrt{\frac{2V_0}{m}}$$

2d) We can say the particle has reached that point when the velocity at that point $x = -d$ is zero.

$$E = \frac{1}{2} m v_0^2 + V(d) = \frac{1}{2} m v^2 = V(-d)$$

$$\frac{1}{2} m v^2 = \frac{V_0}{d^4} ((-d)^2 - d^2)^2 = \frac{V_0}{d^4} (0) = 0 \Rightarrow v = 0$$

The velocity required to reach $x = -d$ is 0.

$$2e) F = -\frac{dV}{dx} = -\frac{4V_0}{d^4} x(x^2 - d^2) \text{ from 2a) calculation}$$

$$= \frac{4V_0}{d^4} x(d^2 - x^2) \quad a = \frac{F}{m} = \frac{4V_0}{md^4} x(d^2 - x^2)$$

$$\vec{F} = \left\langle \frac{4V_0}{d^4} x(d^2 - x^2), 0, 0 \right\rangle$$

$$\nabla \times \vec{F} = (0 - 0)\hat{i} + \left(\frac{\partial}{\partial z} \left(\frac{4V_0}{d^4} x(d^2 - x^2) \right) - 0 \right)\hat{j}$$

$$+ \left(0 - \frac{\partial}{\partial y} \left(\frac{4V_0}{d^4} x(d^2 - x^2) \right) \right)\hat{k} = 0\hat{i} + 0\hat{j} + 0\hat{k} = (0, 0, 0)$$

Since $F_x = \frac{4V_0}{d^4} x(d^2 - x^2)$ is independent of y, z

Therefore, F is conservative due to $\nabla \times F = 0$.

2f. The momentum is $p = m \frac{dx}{dt}$

To check conservation of momentum, we need to find

$$\frac{dp}{dt} = m \frac{dx}{dt^2} = -\nabla V = -\frac{dV}{dx}$$

$$\frac{dp}{dt} = -\frac{d}{dx} \left(\frac{V_0}{d^4} (x^2 - d^2)^2 \right) = \frac{4V_0}{d^4} x(d^2 - x^2)$$

Thus, momentum is not conserved since $\frac{dp}{dt} \neq 0$.

The particle is confined to travel in the x -axis then,

$$L = r \times p = (x, 0, 0) \times (p, 0, 0) = (0, 0, 0)$$

Thus, $\vec{L} = 0$.

We need to check $\frac{dL}{dt}$,

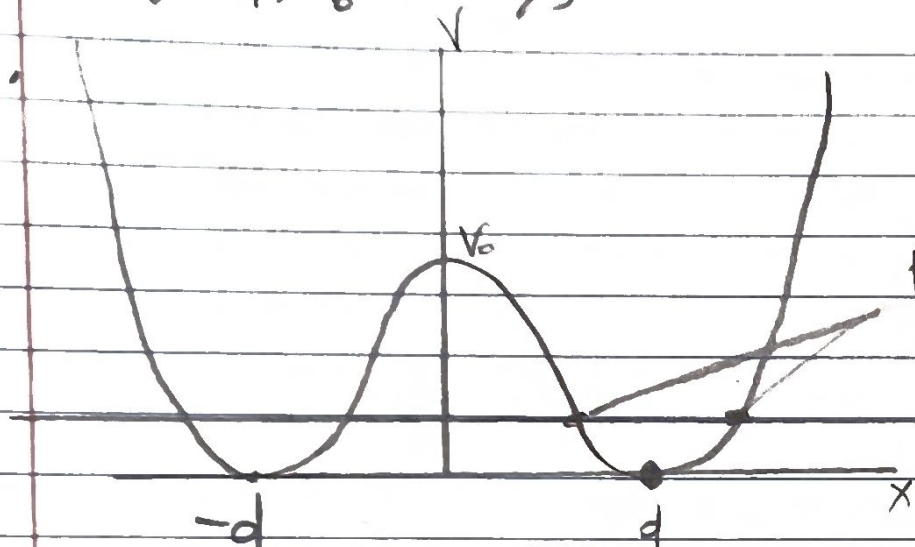
$$\frac{dL}{dt} = r \times F = (x, 0, 0) \times (F_x, 0, 0) = 0$$

Since the force is only along the x -axis.

Therefore, angular momentum is conserved.

$$x_0 = d, v_0 = 0.5 \text{ m/s}$$

2i.

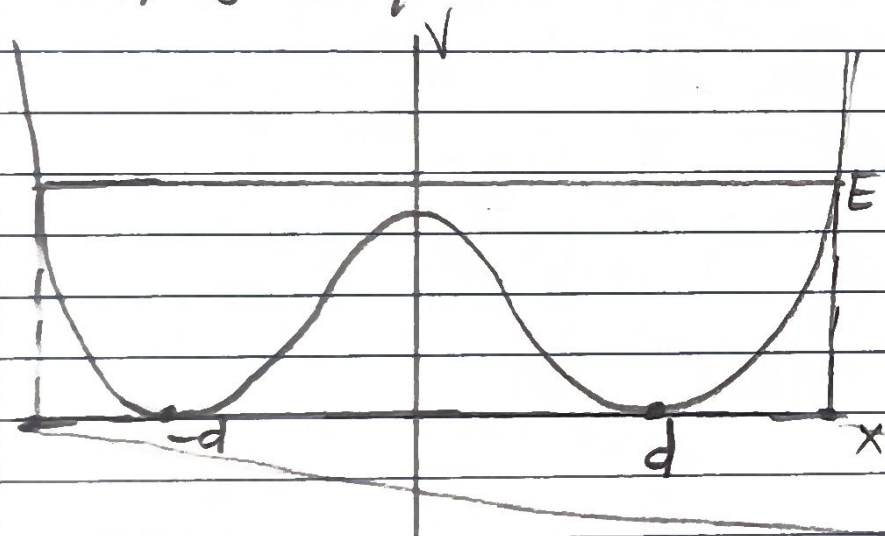


$$V_0 = 1 \text{ J}$$

particle oscillates between these points

$$E = \frac{1}{2}(1)(0.5)^2 + \overbrace{V(d)}^0 = 0.125 \text{ J}$$

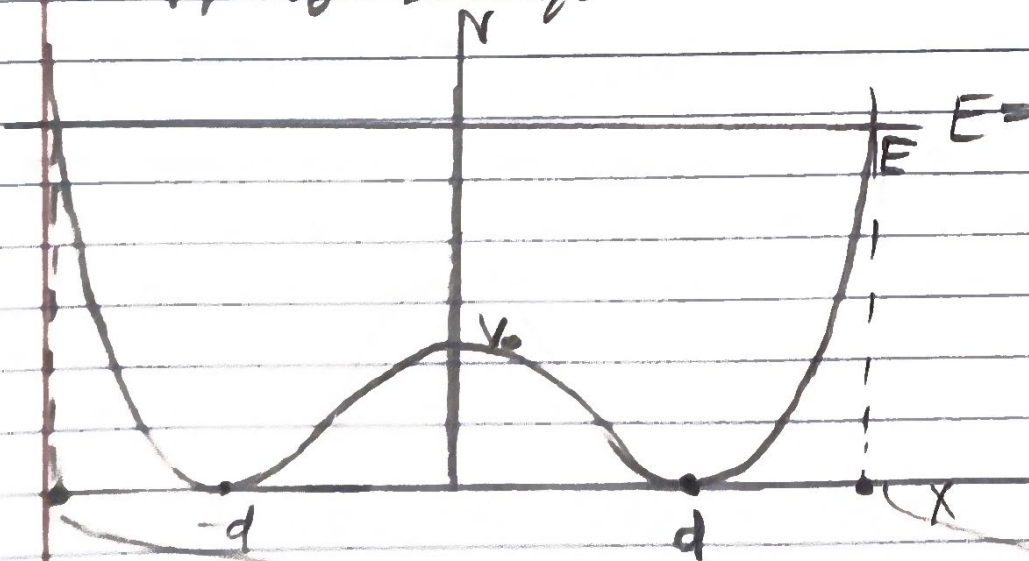
$$x_0 = d, v_0 = 1.5 \text{ m/s}$$



$$E = \frac{1}{2}(1)(1.5)^2 + \overbrace{V(d)}^0 = 1.125 \text{ J}$$

the particle oscillates between these points

$$x_0 = d, v_0 = 2.5 \text{ m/s}$$



$$E = \frac{1}{2}(1)(2.5)^2 + \overbrace{V(d)}^0 = 3.125 \text{ J}$$

The particle oscillates between these points